

Material Safety Data Sheet

May be used to comply with OSHA's Hazard Communication Standard, 29CFR 1910. Standard must be consulted for specific requirements

QUICK IDENTIFIER: ERS-COAL TAR

P/N: 010-0035, 0036

Common Name: COAL TAR PITCH

SECTION 1 -

Manufacturer's

Name Ecology Roof Systems
Address 9821 Olde Eight Road, Unit F

IN CASE OF CHEMICAL EMERGENCY
SPILL, LEAK, FIRE, EXPOSURE, OR

ACCIDENT

City, State, ZIP Northfield, OH 44067

CALL **CHEMTREC** DAY OR NIGHT:

Signature of Person

(800) 424-9300

Responsible for Preparation (Optional)

For Additional Information Call

Date Prepared 05-05-2008

(800) 553-3833

SECTION 2 - HAZARDOUS INGREDIENTS/IDENTITY

Ingredient Name	CAS Number	Weight %
Coal Tar Pitch *	65996-93-2	100%

* Mixture of organic compounds, primarily 3 to 40 ringed polynuclear aromatic hydrocarbons, including some substituted compounds. It is estimated that as many as 4,500+ compounds may be present. Trace impurities and additional material names not listed above may also appear in section 15 towards the end of the MSDS. These materials may be listed for local "Right-To-Know" compliance and for other reasons.

SECTION 3 - HEALTH HAZARDS

Emergency Overview: Vapor can cause moderate to severe irritation of eyes, nose, throat, and respiratory tract. Can cause burning and itching with reddening of the skin, which is accentuated by sunlight. Burning may emit hazardous fumes, which can form flammable or explosive mixtures.

Routs of Entry	Symptoms	Emergency & First Aid Procedures
Eyes	Overexposure to product fumes, vapors or dust can result in irritation and burning. Eye contact with product will result in irritation which in the absence of recommended first aid can result in effects ranging from minor burns to severe corneal injury, including keratitis, conjunctivitis, and corneal abrasion. Contact with heated material may cause thermal burns.	Flush eyes immediately with large amounts of water or olive oil for at least 15 minutes. Call a physician.
Ingestion	Irritation of the gastrointestinal tract followed by nausea and vomiting, abdominal discomfort, rapid pulse, etc.	If conscious, first induce vomiting, then take 2 tablespoons of activated charcoal (USP-drug grade) in water. Do not give anything by mouth to an unconscious person. Get immediate medical attention.
Inhalation	Overexposure to fumes, vapor or dust may result in irritation to respiratory tract. Prolonged exposure in significant excess of permissible air concentrations can result in acute toxic effects, such as coughing, sneezing, headache, dizziness, respiratory difficulty, and convulsions.	Remove immediately to fresh air. If not breathing, give artificial respiration; preferably mouth-to-mouth. If breathing is difficult, give oxygen. Call a physician.
Skin	Contact with skin can result in irritation, which when not washed off or when accentuated by sunlight, can result in minor burns. Contact with heated or molten material can cause severe thermal burns.	For contact with MOLTEN product, do not remove contaminated clothing. Immediately flush skin with large amounts of cold water. If possible, submerge area in cold water. Pack with ice and seek immediate medical attention. For other contact, remove contaminated clothing and wash thoroughly with waterless hand cleaners, olive oil or nonabrasive soap, and water. Avoid

solvents.

SECTION 3 - HEALTH HAZARDS (Continued)

Delayed Effects: Prolonged and repeated skin exposure over many months to years, in the absence of recommended hygiene practices, may lead to changes in skin pigmentation and benign skin growths. In some cases, skin cancer may occur after many years of exposure. These effects appear to be exacerbated by simultaneous exposure to ultraviolet light (sunlight). Long term exposure to coal tar pitch volatiles has been associated with the development of skin, kidney, bladder, scrotum, and lung cancer.

Other: Individuals with chronic respiratory disorders, a history of central nervous system (CNS) functional illness or preexisting skin disorders may be more susceptible to the effects of exposure when working with this material.

Advice to Physician: None.

Ingredients found on one of the OSHA designated carcinogen lists are listed below.

INGREDIENT NAME	NTP STATUS	IARC STATUS	OSHA LIST
Coal Tar Pitch	Known Carcinogen	1 - Known	Carcinogen

SECTION 4 - FIRE & EXPLOSION DATA

Flash Point Method	Pensky-Martens Closed Cup
Flash Point Temperature	374°F min (190°C min)
LEL Flammable Limit	Not Determined
UEL Flammable Limit	Not Determined
Autoignition Temperature	> 750°F (399°C)
Flame Propagation Rate (solids)	Not Applicable
Extinguishing Media	Water fog, carbon dioxide, foam, dry chemicals, sand or steam.
Special Fire Fighting Precautions / Instructions	Wear complete fire service protection equipment, including full-face NIOSH/MSHA approved self-contained breathing apparatus (SCBA). Use water or water spray to cool fire-exposed containers and structures and to protect personnel. Water/fog can control unconfined pitch fires, but may cause frothing or eruption in closed tanks.
Unusual Fire and Explosion Hazards	Sensitive to static discharge. Burning may emit hazardous fumes / vapors which may be in concentrations greater than PEL/TLVs. Coal tar pitch at elevated temperatures may generate vapors that can form flammable / explosive mixtures in the presence of air and a source of ignition. Airborne pitch dust may form explosive mixtures with air. Cloud ignition temperature is 1,310°F (710°C) minimum. Explosive concentration (dust) is 0.035 ounces/ft ³ (1,000 mg / 0.03m ³). Closed containers may explode when exposed to extreme heat. Liquid (molten) pitch at elevated temperatures will sustain combustion.

SECTION 5 - ACCIDENTAL RELEASE MEASURES

In Case of Spill or Other Release: (Always wear recommended personal protective equipment.)

Avoid breathing vapors and contact with skin and eyes. Avoid contact with hot liquid, fumes, or vapors. Avoid sources of ignition (sparks or open flame). Try to stop the source of the leak, if possible, without hazard. Ventilate the area if spill occurs indoors. If hot liquid is spilled, contain by diking, or damming with absorbent solids, such as sand, earth, or other inert material. Use of water spray will aid in solidifying molten material and minimize vapor emissions. Release or spillage of solid pitch can be managed as a coal spillage and recovery made avoiding skin and eye contact. Shovel material into dry, labeled containers and secure cover. Contain runoff of fire control water. Do not allow to enter into sewers, waterways or open bodies of water.

SECTION 5 - ACCIDENTAL RELEASE MEASURES (Continued)

Provide cleanup personnel with appropriate protective clothing. Contaminated materials may need to be handled and managed as RCRA Hazardous Waste and treated before disposal in approved facilities (see Section 13). In cases involving release to the environment in the U.S., report releases to Federal, State, and Local authorities, as required. Due to the concentration of Benzo(a)pyrene in coal tar pitch and a reportable quantity (RQ) of one (1) pound for this compound, CERCLA (Superfund) release of approximately 18 gallons (200 pounds) of coal tar pitch requires National Response Center notification.

Spills and releases may have to be reported to Federal and/or local authorities. See Section 15 regarding reporting requirements.

SECTION 6 - HANDLING AND STORAGE

Normal Handling: (Always wear recommended personal protective equipment).

Avoid prolonged and repeated contact with skin or breathing of dust, fumes, or vapors. Avoid creating aerosols. Avoid contact with molten materials. Wear clothing closed at the neck, long sleeves, and cotton, leather, or non-porous type gloves, (e.g., neoprene, butyl rubber, nitrile, polyvinyl alcohol [PVA], polyvinyl chloride [PVC]).

Storage Recommendations:

Recommended temperature for storage is 122°F (50°C) above the softening point.

SECTION 7 - EXPOSURE CONTROL & PERSONAL PROTECTION

Engineering Controls: Use in areas with adequate natural ventilation or provide sufficient general, local exhaust ventilation in pattern, and volume to maintain concentrations below the recommended PEL/TLV and to maintain areas below flammable vapor or explosive dust concentrations.

Personal Protective Equipment

Skin Protection: Avoid skin contact, whenever possible by using gloves. For exposed skin, use protective creams (e.g., MSA's Fend AE-2, Kerodex 51, Jergens SBS-46). Protect exposed skin from direct sunlight. For outdoor work use approved waterproof sunscreens with a SPF 25 or greater; reapply every 90 minutes while in direct sun.

Eye Protection: Safety glasses, with side shields, goggles, and/or face shield. Chemical splash goggles or face shield are highly recommended when handling molten material. Do not wear contact lenses when handling this material.

Respiratory Protection: Use a NIOSH/MSHA approved respirator with suitable cartridge (organic vapor, high efficiency particulate air filter) as necessary to control exposures to levels below the TLV or PEL. Not required for properly ventilated areas.

Additional Recommendations: Work clothing should be laundered separately from other household clothing. Wash exposed areas thoroughly before eating, drinking, using tobacco products or using a restroom. It is recommended that a complete soap and water shower and/or steam bath be taken at the end of each working day.

Exposure Guidelines

<u>Ingredient Name</u>	<u>ACGIH TLV</u>	<u>OSHA PEL</u>	<u>Other Limit</u>
Coal Tar Pitch Volatiles (CTPV)	02.mg/m ³	0.2mg/m ³	---

SECTION 8 - PHYSICAL AND CHEMICAL PROPERTIES

Physical State	Black solid (at 70°F) or black viscous liquid
Molecular Weight	700 - 900
Chemical Formula	Mixture of organic compounds
Odor	Aromatic
Specific Gravity (H ₂ O = 1)	1.3 ± 0.04 @ 60°F (15.5°C)
Solubility in Water (weight %)	Negligible
pH:	Not Applicable
Boiling Point	> 464°F (240°C)
Melting Point	106 - 147° F (41 - 64°C)
Vapor Pressure	None at 68°F (20°C)
Vapor Density (Air = 1.0)	> 1
Evaporation Rate (Butyl Acetate = 1)	< 1
% Volatiles	Not Determined
Flash Point	374°F (190°C) min

* Flash point method and additional flammability data are found in Section 4.

SECTION 9 - STABILITY AND REACTIVITY

Normally Stable? (Conditions to Avoid): Product stable under normal conditions. Avoid loading or unloading near open flame.

Incompatibilities: Avoid contact with water when confined and in a molten state. Avoid contact with strong oxidizing agents.

Hazardous Decomposition Products: Material does not decompose under normal conditions of use. Upon excessive heating or burning, the material decomposes, and may emit hazardous fumes, or vapors of lower molecular weight compounds, CO₂, CO, NO_x, and SO₂

Hazardous Polymerization: Will not occur.

SECTION 10 - TOXICOLOGICAL INFORMATION

Immediate (Acute) Effects: Refer to Section 3.

Delayed (Sub-chronic and Chronic) Effects: Long term exposure to Coal Tar Pitch Volatiles (CTPV) above the recommended exposure limit, has been associated with the development of skin, kidney, lung, bladder, and scrotum cancer.

Other Data: IARC Group 1 - Sufficient evidence of carcinogenicity in humans. No scientific study supports an association between coal tar pitch exposure and human reproductive hazards. Available data characterizes coal tar pitch as a mutagen.

SECTION 11 - DISPOSAL CONSIDERATIONS

RCRA

Is the unused product an RCRA hazardous waste if discarded? No.
If yes, the RCRA ID number is: Not Applicable.

Other Disposal Considerations:

In the U.S., dispose of the material as required by applicable federal, state, and local regulations.

The information offered here is for the product as shipped. Use and/or alterations to the product such as mixing with other materials may significantly change the characteristics of the material and alter the RCRA classification and the proper disposal method.

SECTION 12 - TRANSPORT INFORMATION

SOLID

US DOT Hazard Class	UN 3077
US DOT ID Number	RQ, Environmentally Hazardous Substance, Solid, N.O.S., (Benzo(a)pyrene, Dibenz(a,h)anthracene), 9, UN 3077, III
	RQ, Other Regulated Substance, Solid, N.O.S., (Benzo(a)pyrene, Dibenz(a,h)anthracene), 9, NA 3077, III
For Domestic Shipments:	Either shipping name
For Marine Shipments:	Use UN 3077, Environmentally Hazardous Substance shipping name.

LIQUID (Transported above flash point)

US DOT Hazard Class	UN 3256
US DOT ID Number	RQ, Elevated Temperature Liquid, Flammable, N.O.S., (Benzo(a)pyrene, Dibenz(a,h)anthracene), 3, UN 3256, III

LIQUID (Transported below flash point)

US DOT Hazard Class	UN 3257
US DOT ID Number	RQ, Elevated Temperature Liquid, Flammable, N.O.S., (Benzo(a)pyrene, Dibenz(a,h)anthracene), 9, UN 3257, III

For additional information on shipping regulations affecting this material, contact the information number found in Sections 1.

SECTION 13 - REGULATORY INFORMATION

Toxic Substances Control Act (TSCA)

TSCA Inventory Status: Listed on EPA's TSCA Inventory
Other TSCA Issues: Substance of unknown or variable composition

SARA Title III/CERCLA

The following substances are considered hazardous by one or more regulatory agencies and were identified in "typical" coal tar pitch samples at concentrations greater than 0.01 percent by weight. "Reportable Quantities" (RQs) and/or "Threshold Planning Quantities" (TPQs) exist for the following ingredients.

INGREDIENT NAME	CAS #	SARA/CERCLA RQ (lb)	SARA EHS TPQ (lb)
Acenaphthene	83-32-9	100	
Anthracene	120-12-7	5,000	
Benzo(a)anthracene	56-55-3	10	
Benzo(c)acridine	225-51-4	100	
Benzo(b)fluoranthene	205-99-2	1	
Benzo(k)fluoranthene	207-08-9	5,000	
Benzo(j)fluoranthene	205-82-3	- - -	
Benzo(g,h,i)perylene	191-24-2	5,000	
Benzo(a)pyrene	50-32-8	1	
Benzo(e)pyrene	192-97-2	- - -	
Chrysene	218-01-9	100	
Dibenz(a,h)anthracene	53-70-3	1	
Dibenzofuran	132-64-9	100	
Fluoranthene	206-44-0	100	
Fluorene	86-73-7	5,000	
Indeno(1,2,3-cd)pyrene	193-39-5	100	
2-Methylnaphthalene	91-57-6	- - -	
Naphthalene	91-20-3	100	
Phenanthrene	85-01-8	5,000	
Pyrene	129-00-0	5,000	1,000/10,000*

* See 40 CFR 355.30(2)(i)

SECTION 13 - REGULATORY INFORMATION (Continued)

Spills or releases resulting in the loss of any ingredient at or above RQ requires immediate notification to the National Response Center, (800) 424-8802, and to your Local Emergency Planning Committee.

Section 311 Hazard Class: Immediate, Delayed, Fire

SARA 313 Toxic Chemicals

The following ingredients are SARA 313 "Toxic Chemicals". Also see Section 2.

Ingredient Name	CAS #	% Weight
* Benzo(a)anthracene	56-55-3	0.7
* Benzo(a)pyrene	50-32-8	0.4
* Benzo(b)fluoranthene	205-99-2	0.8, Includes(k)
* Benzo(k)fluoranthene	207-08-9	Included above
* Benzo(j)fluoranthene	205-82-3	0.2
* Benzo(g,h,i)perylene	191-24-2	0.2
* Chrysene	218-01-9	0.2
* Indeno(1,2,3-cd)pyrene	193-39-5	0.2
Phenanthrene	85-01-8	1.8
Polycyclic Aromatic Compounds (20 compounds)	-	3.2

* Also included in "polycyclic aromatic compound" category.

State Right-To-Know

In addition to the ingredients found in Section 2, the following are listed for state right-to-know purposes:

Ingredient Name	% Weight	Comment
No ingredients listed in this section		

Additional Regulatory Information: California Proposition 65 material - Contains chemicals known to the State of California to cause cancer, birth defects, & other reproductive harm.